CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (CL) Examination

December 2012

CL101.01 Fundamentals of Civil Engineering

Date: 11.12.2012, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) (1) Enlist the properties of good brick. [03]

(2) Write uses of cement. [03]

(b) Which branch of civil engineering deals with soil investigation and foundation design? [01]

Q - 2 (a) Write short note on BOT projects for highways. [06]

(b) Explain detail classification of surveying based on instrument used. [04]

(c) Give the functions of the following; [04]

Weather shed, Plinth, Slab, Footing.

OR

Q - 2 (a) Enlist the types of roads based on Nagpur road plan. Explain each in detail. [06]

(b) Explain two fundamental principles of Surveying. [04]

(c) Write short note on privacy and circulation. [04]

Q - 3 Refer **Figure-1** and draw section at A-A. Use levels and schedule of opening given below the figure. Assume suitable scale. [14]

OR

Q - 3 Refer **Figure-1** and draw elevation from the side where arrow is given. Use levels and schedule of opening given below the figure. Assume suitable scale. [14]

SECTION – II

Q - 4 (a) Explain with neat sketch: Hydrological Cycle. [05]

(b) Define Confined aquifer and Unconfined aquifer. [02]

Q - 5 (a) Nine perpendicular offsets were taken from a survey line at an interval of 10 m to a curved boundary. Calculate the area by Simpson's rule. [05]

Distance(m)	0	10	20	30	40	50	60	70	80
Offsets (m)	1.2	1.8	2.1	3.5	1.7	2.8	1.66	0.9	1.84

(b) (1) Define **Any Two**: Local attraction, Dip of needle and Back bearing of a survey line. **[05]**

(2) Convert following whole circle bearings to quadrantal bearings:

(i) $25^\circ 30'$ (ii) 135° (iii) $281^\circ 30'$

(c) Draw neat sketch of 20 m long chain & explain procedure to measure a distance using the same chain. **[04]**

OR

Q - 5 (a) Explain the construction and working of Planimeter with a neat sketch. **[05]**

(b) The following bearings were observed in running a closed traverse ABCDE. Calculate the interior angles of the traverse and apply necessary checks. **[05]**

Line	Fore Bearing
AB	140°
BC	81°
CD	340°
DE	289°
EA	232°

(c) What is ranging? Explain reciprocal ranging with neat sketch. **[04]**

Q - 6 (a) Define the following **(Any Six)**: Mean sea level, GTS Bench mark, Reduced level, Bench mark, Line of collimation, Height of instrument and Change point **[06]**

(b) Differentiate between Prismatic Compass and Surveyor's Compass. **[05]**

(c) Enlist the types of water losses. Explain any one of them in detail. **[03]**

OR

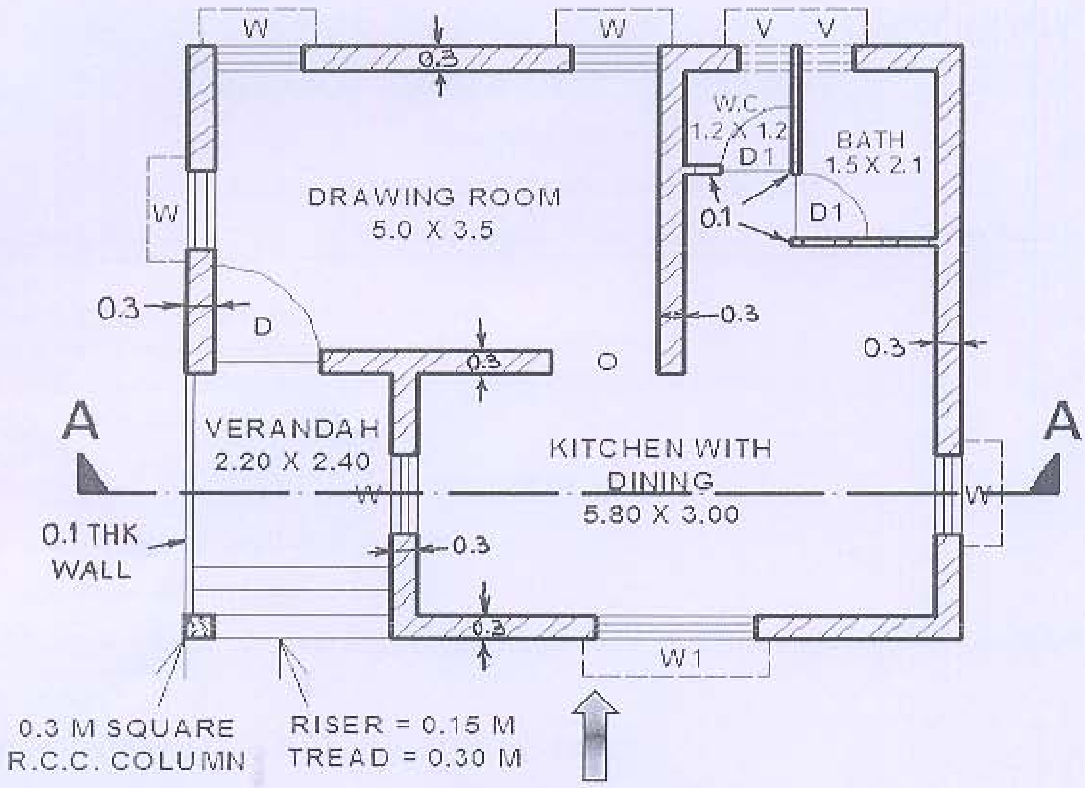
Q - 6 (a) The following consecutive readings were taken with a dumpy level: 3.875, 3.630, 2.865, 1.945, 0.920, 3.165, 2.860, 1.895, 2.125, 0.965 and 0.785. **[06]**

The instrument was shifted after fifth and eighth readings. The first reading was taken on Bench mark of 260.865 m RL. Use Rise and Fall method and Height of Instrument method to calculate the reduced levels of all points and apply usual checks.

(b) Explain temporary adjustments of a dumpy level, in detail. **[05]**

(c) Write short note on 'Water Conveyance System'. **[03]**

****



NOTE:- ALL DIMENSIONS ARE IN METER.

FIGURE - 1

SCHEDULE OF OPENING		
SR.NO.	DESCRIPTION - SIZE	NO(S)
1.	DOOR 'D' - 1.5 X 2.0	1
2.	DOOR 'D1' - 0.8 X 2.0	2
3.	OPENING 'O' - 1.0 X 2.0	1
4.	WINDOW 'W' - 1.0 X 1.0	5
5.	WINDOW 'W1' - 1.8 X 1.0	1
6.	VENTILATOR 'V' - 0.6 X 0.6	2

REQUIRED LEVELS:-

Plinth Level = 0.60 m above Ground Level

Lintel Level = 2.00 m above Plinth Level

Slab Bottom = 3.00 m above Plinth Level

Slab Thickness = 0.10 m thick

Parapet Height = 0.90 m above Slab Top.